AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2012

Computer Science Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

A mathematics teacher wants to calculate the area and circumference of a circle.

He takes the diameter of the circle as input. Design a solution by writing an algorithm.

Area = π radius^2, Circumference = 2π radius, where π = 3.14.

Q.2. (Total 5 Marks)

State the purpose of the following GW BASIC commands:

i. AUTO _____

_____

ii. DELETE 50 -90 _____

_____

iii. LIST _____

_____

iv. LOAD _____

_____

v. NEW _____

_____

Q.3. (Total 5 Marks)

Consider the following GW BASIC program.

```
10 LET N = 5
20 FOR A = 1 TO 10 Step 2
30 LET T = N*A
40 PRINT T
50 NEXT A
60 END
```

a. Which one is the control variable in the given GW BASIC program? (1 Mark)

_____

_____

b. Convert the FOR NEXT loop into WHILE WEND loop in the given GW BASIC program. (4 Marks)

_____

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Q.4. (Total 4 Marks)

Declare the following array variables using GW BASIC.

i.	Type: numeric	Size: 15	Name: A	
ii.	Type: string	Size: 8	Name: B	
iii.	Type: string	Number of rows: 4	Number of columns: 3	Name: C
iv.	Calculate the size of array C.			

Q.5. (Total 3 Marks)

Using GW BASIC statement, draw a circle of radius 50, with the coordinates of centre, x-centre = 100, y-centre = 150 in high resolution graphics mode.

Q.6. (Total 3 Marks)

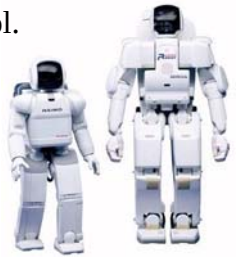
Complete the following truth table for Boolean operators NAND and NOR.

A	B	A NAND B	A NOR B
0	**0**	**1**	**1**
0	**1**		
1	**0**		
1	**1**		

Q.7. (Total 8 Marks)

EITHER

a. The word ‘robot’ comes from the Czech word 'robota' which means 'forced work or labour'. Today, we use the word ‘robot’ to mean any man-made machine that can perform work or other actions normally performed by humans, either automatically or by remote control.



i. Write THREE applications of robots in the car industry.
ii. Write THREE general advantages of using robots.
iii. Write TWO general limitations of using robots.

OR

b. The advance in computer technology are bringing revolution in our lives. Discuss FOUR positive impacts of computer technology in the following fields.

i. Teaching and Learning ii. Business

Q.8. (Total 7 Marks)

EITHER

a. Write a GW BASIC program to read an array of 12 numeric elements with INPUT statements which calculate and print the product of these numbers.

OR

b. Write a GW BASIC program to calculate and print the roots of quadratic equation.
Take a, b, c as input using READ DATA statement.

Root1 = $\frac{-b+\sqrt{b^2-4ac}}{2a}$, Root2 = $\frac{-b-\sqrt{b^2-4ac}}{2a}$

Also find the nature of roots. If $b^2-4ac < 0$ then roots are imaginary, otherwise roots are real.

END OF PAPER

Please use this page for rough work

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